FASAL SURAKSHA

AI Micro-Crop Diagnostic, Yield Forecast & PMFBY Verification Engine

Issue Date: 11 Oct 2026, 21:37 | Card ID: FSC-734836

Verified via Grad-CAM XAI Engine

Target Crop:	Diagnosis:	Severity:	AI Confidence:
Potato	**Early Blight**	**4.2% (Moderate)**	**95.0%**

Original Leaf Capture

[Camera Scan Input]

Grad-CAM AI Heatmap

[Grad-CAM Overlay]

Yield Impact Simulation (XGBoost)

Potential Healthy Yield: 22.0 Tons/Acre

Forecasted Harvest: **14.80 Tons/Acre**

Projected Deficit / Loss: **32.7%**

Model simulates crop age (65d), seasonal precipitation, soil pH (6.5), and pathological leaf necrosis index calibrated for agricultural zones.

Prescribed Treatment Plan (ICAR / Agronomic Standards)

Category	Prescribed Formulation	Concentration & Tank Dose	Frequency & Application
Organic / Bio	**Neem Oil 10,000 PPM + Trichoderma harzianum** Bio-fungicide protective formulation	3.0 mL / Liter water - 45.0 mL for 15L tank	Every 7 days. Spray at first sign of foliar lesions.
Chemical	**Mancozeb 75% WP (Dithane M-45)** Systemic curative fungicide	- 2.5 g / Liter water - 37.5 g for 15L tank	Every 10 days. Uniform foliar application covering both canopy surfaces.

PMFBY Crop Insurance Evidence & Verification Dossier

Claim Token:	Farmer / KCC ID:	Claim Eligibility:	Est. Indemnity:
PMFBY-2026-X89	**KCC-FMR-98214**	**YES (Admissible)**	**INR 22,425.00**
Audit Priority:	Surveyor Directive:	Metadata Integrity:	Scheme Reference:
TIER-2: FAST-TRACK DIGITAL ADJUDICATION	Damage crosses 20% statutory threshold. Approved.	Tamper Score: 96% | Geo: Verified	PMFBY Localized Calamities Norm

Digitally generated by FASAL SURAKSHA AI Platform. Complies with PMFBY automated claims adjudication guidelines and ICAR crop protection standards. For re-scan or dispute redressal, visit nearest Krishi Vigyan Kendra (KVK).